Structured questions

|!|BCELA0300000001|!| ★★

A scientist investigated the evolutionary relationship among humans and two other species (X and Y). She collected DNA samples from the species for the following analysis:

1. The DNA samples were heated and the DNA molecules separated to single strands.
2. The single DNA strands from the three species were mixed and allowed to cool. As the strands cooled, they rejoined to form double helices. Some helices were hybrid (the two strands came from different species).
3. The hybrid helices were heated again. The temperature at which the strands separate was recorded in the table below.

The two DNA strands of a helix came from		**Temperature required to separate the two strands of a helix**
Humans	Humans	94 ^{o}C
Humans	Species X	92.4 ^{o}C
Humans	Species Y	90.4 ^{o}C

(a) Using your knowledge of DNA structure, explain why the less closely related the species, the lower the separation temperature. (3 marks)

(b) Based on the information given, construct an evolutionary tree to represent the evolutionary relationship among humans, species X and Y. (2 marks)

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(c) Explain why scientists can also compare protein structure to investigate the evolutionary relationship between different species. (2 marks)

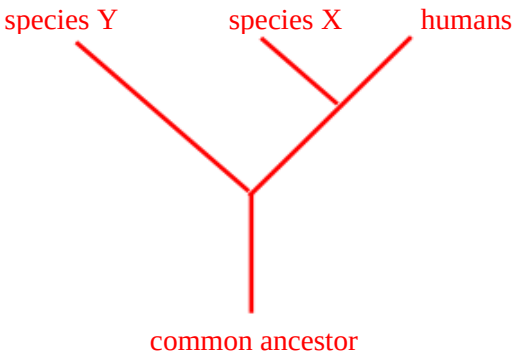
##

(a) The less closely related the species, the more differences are found in their DNA base sequences. (1)

When the single strands from the less closely related species join to form double helices, fewer complementary base pairs form. (1)

Therefore, less energy is required to break the hydrogen bonds that hold the base pairs. (1)

(b)



Labelling (1)

Correct tree (1)

(c) More closely related species have more similarities in the amino acid sequence in the same protein (e.g. haemoglobin). (1)

A higher similarity in amino acid sequence suggests a higher similarity in the DNA base sequence. (1)

(7 marks)

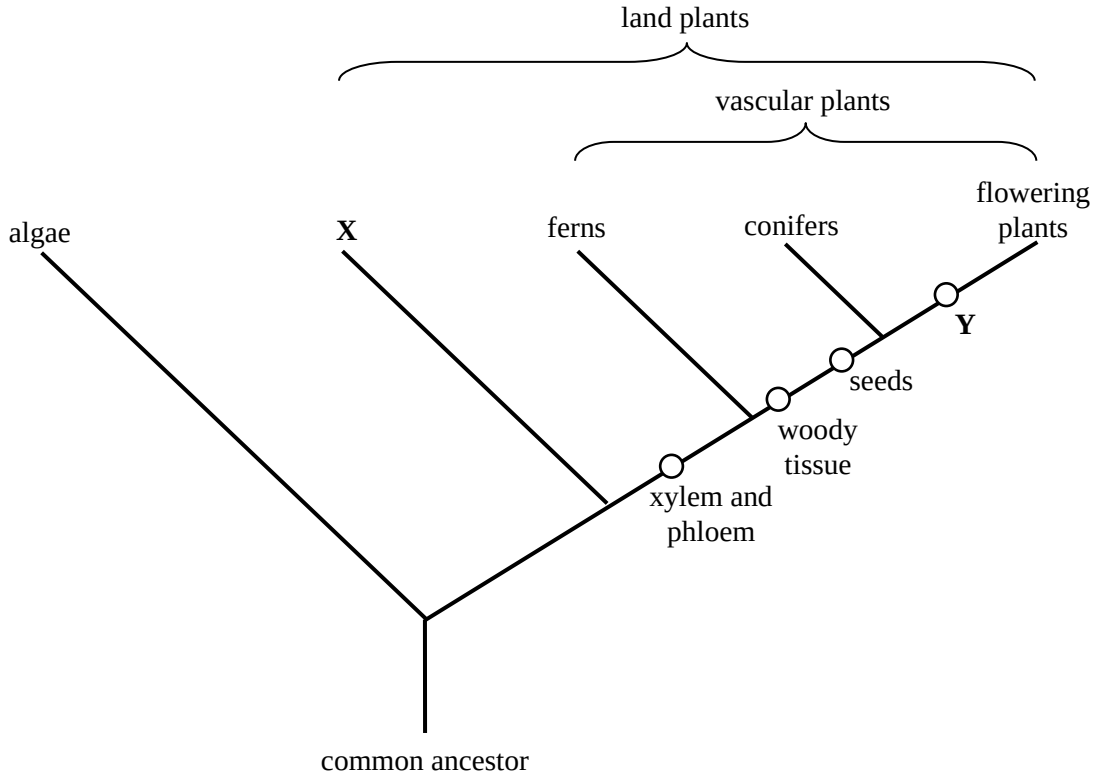
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Analysing data	1
	Presenting information	1
(c)	Understanding basic concepts	2

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|!|BCELB0300000002|!| ★ **Cross topic**

The diagram below shows possible evolutionary relationship among plant-like protists and different groups of land plants.



(a) Give ***one*** similarity and ***one*** difference between algae and land plants. (2 marks)

(b) Identify plant group X. (1 mark)

(c) Give ***one*** example of characteristic (Y) found in flowering plants but not in conifers. (1 mark)

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(d) Scientists study fossil records of organisms to deduce the evolution of organisms.

(i) Describe the commonest way of fossil formation. (4 marks)

(ii) Give ***two*** limitations of using the fossil record as evidence for evolution. (2 marks)

##

(a) Similarity:
Algae and green plants have photosynthetic pigments that are required for carrying out photosynthesis to make food / other reasonable answer (1)
Difference:
Some algae are unicellular while all plants are multicellular / other reasonable answer (1)

(b) Mosses (1)

(c) Flowering plants produce flowers for reproduction / other reasonable answer (1)

(d) (i) After an organism died under water, its dead body was buried in sand and mud. (1)
Its soft parts decayed over time, leaving only the hard parts such as the skeleton. (1)
Under the pressure from the upper layers of sediment, sand and mud around the skeleton turned to rock over millions of years. (1)
Mineral salts deposited in the skeleton and hardened it into a fossil. (1)

(ii) Any ***two*** of the following: (2)

- Some organisms have soft bodies which are difficult to form fossils.
- Some organisms may live in an environment where formation of fossils is not feasible.

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- Since natural processes such as erosion and tectonic movements may destroy fossils, most fossils are found in fragments.
- Some fossils are formed in inaccessible areas.

(Or other reasonable answers)

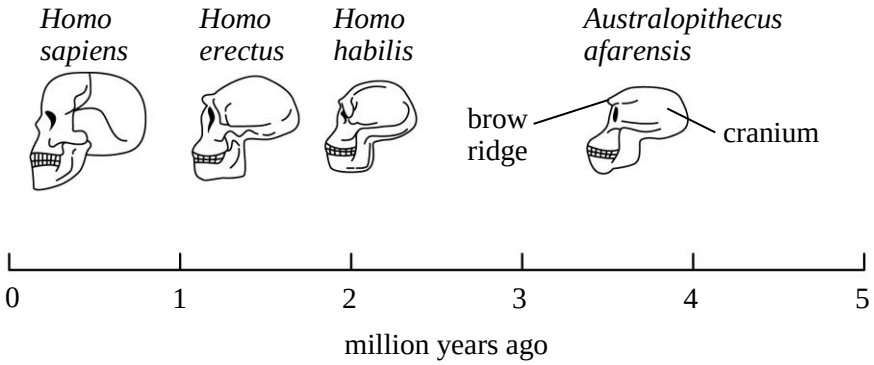
(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)(i)	Understanding basic concepts	4
(c)(ii)	Understanding basic concepts	2

|!|BCELA0300000003|!| ★★

Scientists study the fossils of possible ancestors of modern humans (*Homo sapien*) and determine their time of existence, as shown in the timeline below.



(a) Describe how the skull may become a fossil. (3 marks)

(b) How can scientists determine the age of fossils? (1 mark)

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(c) With reference to the given diagrams, describe ***two*** structural differences between the skull of *Australopithecus afarensis* and *Homo sapiens*. (2 marks)

(d) What clues in the fossils of human ancestral species suggest that they walk with two legs? Give ***one*** example. (1 mark)

(e) (i) Besides fossils, scientists compare the DNA base sequences of different species to deduce their evolutionary relationship. Explain the principle of this method. (3 marks)

(ii) State ***one*** limitation of this method. (1 mark)

##

(a) The dead body was buried in sand and mud. (1)
The soft tissues decayed over time, leaving only the skeleton. (1)
Under pressure from the upper layers of sediments over millions of years, the bone tissues in the skull were replaced by mineral salts and became fossils. (1)

(b) Use radioisotope dating to estimate the age of rocks and fossils (1)

(c) Any ***two*** of the following: (2)

- The cranial capacity of the skull is larger in *Homo sapiens*.
- The brow ridge is less prominent in *Homo sapiens*.
- The jaws are less protruding in *Homo sapiens*.

(Or other reasonable answers)

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(d) Any ***one*** of the following: (1)

- By observing the fossils of their footprints
- By observing the structure of their feet / hip joints / spines

(Or other reasonable answers)

(e) (i) The base sequences of DNA is the genetic code that directs protein synthesis, therefore it determines the characteristics of organisms. (1)

Organisms with a common ancestor usually share many similar features because of a higher similarity in the base sequences of their DNA. (1)

Hence, it is assumed that organisms that share more similarities in the base sequence of DNA have a closer evolutionary relationship than those with fewer similarities. (1)

(ii) Remains or fossils of extinct species contain no DNA. / DNA of extinct species has become degraded. (1)

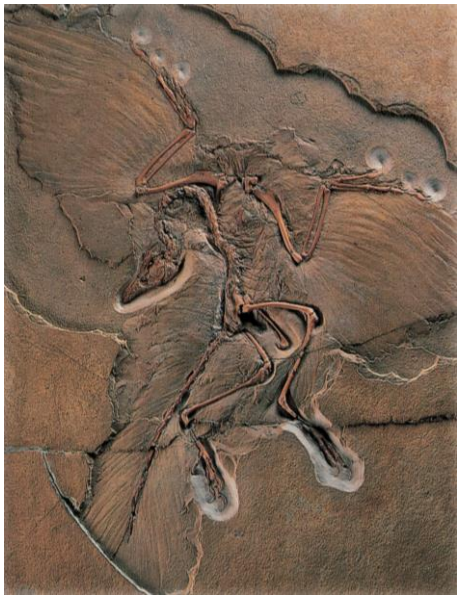
(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	1
(c)	Analysing data	2
(d)	Applying concepts	1
(e)(i)	Understanding basic concepts	3
(e)(ii)	Applying concepts	1

|!|BCELB0300000004|!| ★

The following photograph shows a fossil of an extinct bird-like animal called *Archaeopteryx*.



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(a) Suggest how the fossil of *Archaeopteryx* was formed. (3 marks)

(b) Scientists have found other fossils of the ancestors of modern birds, but the fossil record is incomplete. Suggest ***two*** reasons why there are gaps in the fossil record. (2 marks)

(c) According to the photograph, suggest why this species was the intermediate form between reptiles and birds. (2 marks)

##

(a) The dead body was buried in sand and mud. (1)
The soft tissues decayed over time, leaving only the skeleton. (1)
Under pressure from the upper layers of sediments over millions of years, the bone tissues were replaced by mineral salts and became fossils. (1)

(b) Any ***two*** of the following: (2)

- Some organisms died in conditions that were not favourable for fossil formation.
- Natural processes such as erosion and movements of the Earth's crust may damage fossils, leaving only incomplete fragments.
- Some fossils are trapped in inaccessible areas.

(Or other reasonable answers)

(c) This species had some features of reptiles, e.g. long bony tail / teeth. (1)
This species had some features of birds, e.g. feathers / wings. (1)

(7 marks)

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##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	2
(c)	Applying concepts	2

|!|BCELA0300000005|!| ★★

The haemoglobin molecule in mammals is composed of four polypeptide chains. Scientists analysed the amino acid sequence in one polypeptide chain of haemoglobin found in four species of apes. Differences in the amino acid sequence occur at three positions in the polypeptide chain as shown in the table below.

Species	Position 87	Position 104	Position 125
Bonobo	T	R	P
Gorilla	T	K	P
Orang-utan	K	R	Q
Chimpanzee	T	R	P

(a) Which two species of apes in the table share the closest evolutionary relationship? Explain your answer. (3 marks)

(b) Construct an evolutionary tree to show the evolutionary relationship among the four species. (2 marks)

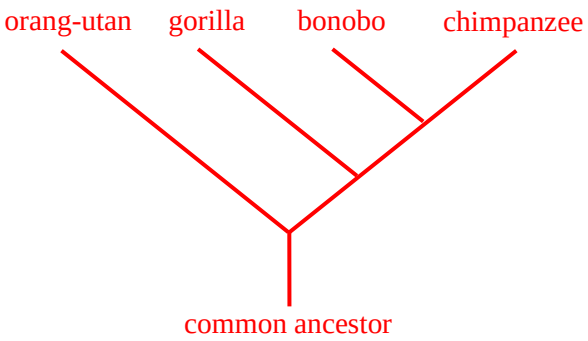
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(c) The scientists want to further study the evolutionary relationship among the four species of apes and humans. Suggest another type of biochemical data suitable for this analysis and explain the principle. (4 marks)

##

(a) The bonobo and the chimpanzee share the closest evolutionary relationship. (1)
Their amino acid sequences of the polypeptide chain being analysed are identical. (1)
The fewer the number of amino acid differences of a protein found in different organisms, the more closely related the organisms are. (1)

(b)



Correct tree (1)
Correct labelling (1)

(c) Comparing the DNA base sequences of the four species of apes and humans (1)
The base sequences of DNA is the genetic code that directs protein synthesis, therefore it determines the characteristics of organisms. (1)
Organisms with a common ancestor usually share many similar features because of a higher similarity in the base sequences of their DNA. (1)
Hence, it is assumed that organisms that share more similarities in the base sequence of DNA have a closer evolutionary relationship than those with fewer similarities. (1)

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(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Presenting information	2
(c)	Applying concepts	4

|!|BCELA0300000006|!| ★★

In comparative biochemistry, scientists can compare the antibodies produced by different organisms in order to find out their phylogenetic relationship. In a study, serum containing human blood proteins was injected into a rabbit. Serum containing the antibodies against human blood proteins was then obtained from the rabbit and added to sera obtained from five different animals. The percentage of precipitation in each mixture was measured and recorded in the table below.

Mammal	Percentage of precipitation (%)
Human	100
Gibbon	80
Spider monkey	59
Chimpanzee	97
Baboon	75

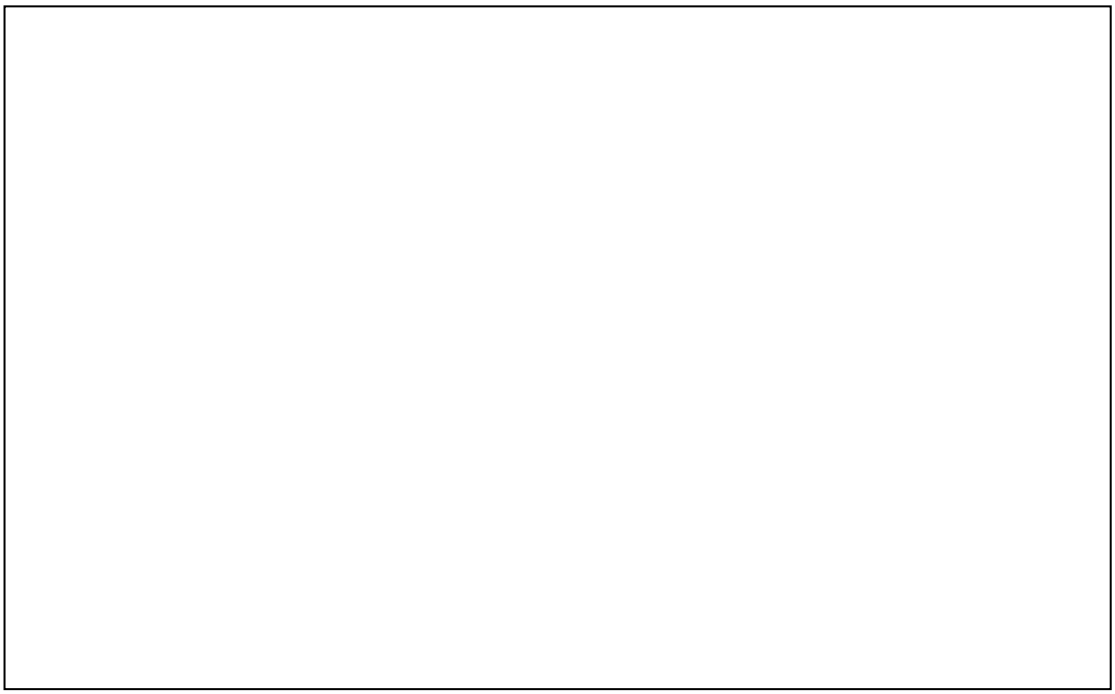
(a) (i) What is an antibody? (2 marks)

(ii) Which type of cells in mammals produce antibodies? (1 mark)

(b) Explain how phylogenetic relationships of these animals may be deduced by comparing the amount of precipitate produced. (4 marks)

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(c) Based on the above information, construct an evolutionary tree of the five animals assuming that they were descended from a common ancestor. (2 marks)



##

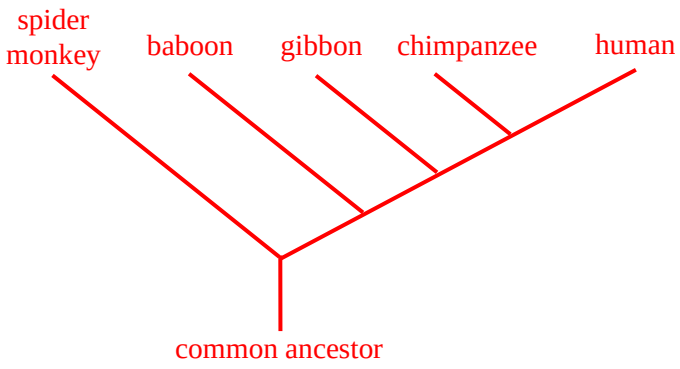
(a) (i) An antibody is a Y-shaped protein molecule (1)
which has antigen-binding sites that bind to specific antigens. (1)

(ii) plasma cells (1)

(b) Antibodies produced by the rabbit are specific to the proteins that trigger their production. Precipitation occurs if these antibodies meet the same protein again. (1)
The degree of precipitation thus depends on how many of the animal's blood proteins are the same as / similar to those of humans. (1)
Since proteins are encoded by genes, this further reflects on how many of the animal's blood protein genes are the same as / similar to those in humans. (1)
The greater the number of genes in common with human, the closer its phylogenetic relationship is with humans. (1)

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(c)



Correct tree (1)

Correct labelling (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	1
(b)	Applying concepts	4
(c)	Presenting information	2

|!|BCELA0300000007|!| ★★

Scientists suggest that humans might have descended from a species of apes. The species most closely related to humans are the chimpanzee and the gorilla. Comparative cytology and biochemistry provide evidence for the evolutionary relationship among these three primates, as listed in the table below.

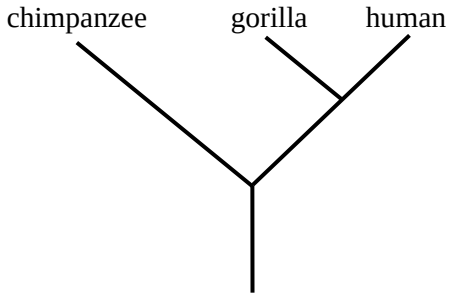
Evidence	Human	Gorilla	Chimpanzee
Chromosome number	46	48	48
Amino acid sequences of plasma proteins	Same as both gorillas and chimpanzees	Same as both humans and chimpanzees	Same as both humans and gorillas
Amino acid sequence of myoglobin	Differs by one amino acid from those of gorillas and chimpanzees	Same as chimpanzees	Same as gorillas
Amino acid sequence of haemoglobin	Same as chimpanzees	Differs by one amino acid from those of humans and chimpanzees	Same as humans

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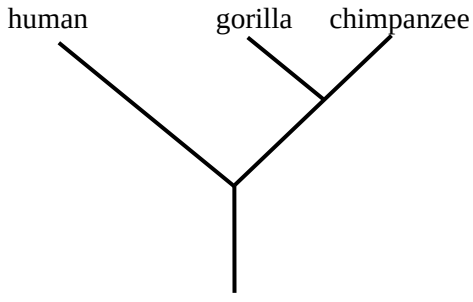
(a) Based on the evidence from analysis of the amino acid sequence of haemoglobin, construct an evolutionary tree for humans, gorillas and chimpanzees. (2 marks)

(b) Jack and Katy each draws an evolutionary tree for humans, gorillas and chimpanzees based on the information in the table.

Jack's evolutionary tree



Katy's evolutionary tree



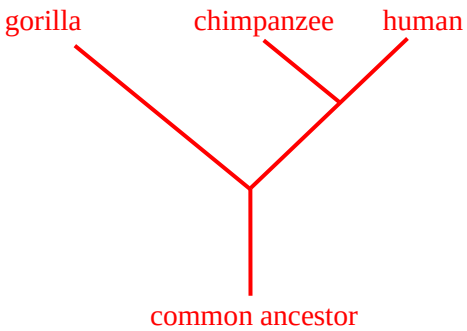
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Evaluate the two evolutionary trees based on the information in the table. (3 marks)

(c) The information in the table shows conflicting evidence in reflecting the evolutionary relationship among the three primates. Suggest another type of biochemical data suitable for determining evolutionary relationship among the three primates and explain the principle. (4 marks)

##

(a)



Correct tree (1)

Correct labelling (1)

(b) Jack's evolutionary tree is not supported by any of the evidence in the table. (1)

Katy's evolutionary tree is supported by evidence from chromosome number and myoglobin analysis. (1, 1)

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(c) Comparing the DNA base sequences of the three species (1)

The base sequences of DNA is the genetic code that directs protein synthesis, therefore it determines the characteristics of organisms. (1)

Organisms with a common ancestor usually share many similar features because of a higher similarity in the base sequences of their DNA. (1)

Hence, it is assumed that organisms that share more similarities in the base sequence of DNA have a closer evolutionary relationship than those with fewer similarities. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Presenting information	2
(b)	Analysing data	3
(c)	Applying concepts	4

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